



2IA Recap Problems

It is that time of the year ...



Recap problems

1. In each case determine whether the statement is true or false. **You will get zero marks if you do not provide an argument supporting your answer.**
 - (a) $\mathbb{Z}$ and $\mathbb{Q}$ are isomorphic.
 - (b) The cyclic group $\mathbb{Z}_n$ has exactly n generators.
 - (c) The intersection of any two subgroups of a group G can never be the empty set.
 - (d) $\mathbb{Z}_2$ is a subgroup of $\mathbb{Z}_4$.
 - (e) Every two cyclic groups of the same order are isomorphic.
 - (f) There exists group G that is isomorphic to a proper subgroup of itself.
2. Let G be a group. Prove that G is abelian if and only if $(ab)^2 = a^2b^2$ for all $a, b \in G$.
3. Let G be a group and $z \in Z(G)$. Write G^z to denote the set G with the operation $a * b = abz^{-1}$. Prove the following assertions:
 - (a) G^z is a group.
 - (b) $G \cong G^z$.
4. Let G be a group with order $|G| < 400$. Suppose that G has subgroups of orders 45 and 75 and find $|G|$.
5. Let H and K be subgroups of a group G such that $|H| = p$, where p is a prime number. Prove that either $H \subseteq K$ or $H \cap K = \{1\}$.
6. Let G be a group, H, K subgroups of G and $a, b \in G$. Prove that either $Ha \cap Kb = \emptyset$ or $Ha \cap Kb = (H \cap K)c$ for some $c \in G$.
7. Let K be a subgroup of G of index 2.
 - (a) Determine the right cosets of K in G .
 - (b) Prove that K is normal in G .
 - (c) If $a \in G \setminus K$ and $b \in G \setminus K$, prove that $ab \in K$.

8. Let H be a subgroup of a group G such that if $a, b \in G$ satisfy $Ha \neq Hb$, then $aH \neq bH$. Prove that H is normal in G .

9. Let G be a group and $\varphi : G \rightarrow G$ an automorphism of G ; that is, a group homomorphism that is bijective.
- (a) Prove that the subset $H = \{g \in G \mid \varphi(g) = g\}$ consisting of all the elements of G fixed by φ is a subgroup of G .
 - (b) Let G be finite now. Suppose that $\varphi : G \rightarrow G$ fixes more than half of the elements of G and show that φ is the identity automorphism; that is, $\varphi(g) = g$ for all $g \in G$.